16-bit ALU Design in VHDL

Last time, I introduced the N-bit adder design in Verilog, which is a part of a 16-bit ALU design I will present today. The 16-bit ALU is a core combinational component of the processing unit in the coprocessor I introduced in the previous post.

Full VHDL code for 16-bit ALU together with testbench will be presented in this VHDL project.

The instruction set of the 16-bit ALU is as follows:
1. ADD: ABUS + BBUS -> ALUOUT
2. SUB: ABUS - BBUS -> ALUOUT
3. AND: ABUS & BBUS -> ALUOUT
4. OR: ABUS | BBUS -> ALUOUT
5. XOR: ABUS ^ BBUS -> ALUOUT
6. NOT: ~ABUS -> ALUOUT
7. MOV: ABUS -> ALUOUT
The addition/ subtraction of the 16-bit ALU is designed and implemented using the Verilog N-bit Adder.
VHDL code for 16-bit ALU:

```
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for 16-bit ALU
-- Top level VHDL code for 16-bit ALU
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
-- 16-bit ALU
entity ALU is
 port (
   ABUS: in std_logic_vector(15 downto 0); -- ABUS data input of the 16-bit ALU
   BBUS: in std_logic_vector(15 downto 0); -- BBUS data input of the 16-bit ALU
   ALUctrl: in std_logic_vector(3 downto 0); -- ALUctrl control input of the 16-bi
   ALUOUT: out std_logic_vector(15 downto 0)  -- 16-bit data output of the 16-bit
   );
end ALU;

architecture Behavioral of ALU is
-- N-bit Adder in Verilog
component N_bit_adder is
generic (
    N: integer:=32
   );
port( input1: in std_logic_vector(N-1 downto 0);
  input2: in std_logic_vector(N-1 downto 0);
  answer: out std_logic_vector(N-1 downto 0)
 );
end component N_bit_adder;
signal BBUS_not: std_logic_vector(16-1 downto 0);
signal tmp_out1: std_logic_vector(16-1 downto 0);
signal tmp_out2: std_logic_vector(16-1 downto 0);
```

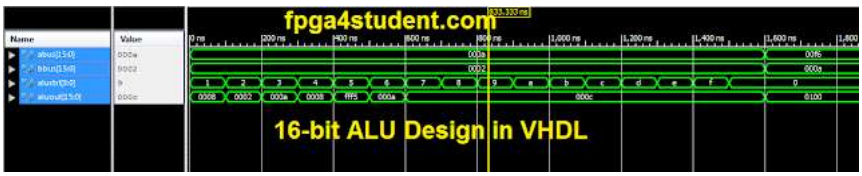
```
signal tmp: std_logic_vector(16-1 downto 0);
begin
-- instantiate Verilog N-bit Adder in VHDL code
u1_N_bit_adder: N_bit_adder generic map ( N => 16) -- ABUS + BBUS
     port map( input1 => ABUS, input2 => BBUS,answer => tmp_out1 );
u2_N_bit_adder: N_bit_adder generic map ( N => 16) -- ABUS + (~BBUS)
     port map( input1 => ABUS, input2 => BBUS_not,answer => tmp_out2 );
u3_N_bit_adder: N_bit_adder generic map ( N => 16) -- ABUS + (~BBUS) + 1 = ABUS -
     port map( input1 => tmp_out2, input2 => x"0001",answer => tmp );
BBUS_not <= not BBUS;
-- Other instructions of the 16-bit ALU in VHDL
process(ALUctrl,ABUS,BBUS,tmp_out1,tmp)
begin
case(ALUctrl) is
 when "0000" =>  ALUOUT <= tmp_out1;  -- ADD
 when "0001" =>  ALUOUT <= tmp ;-- SUB
 when "0010" =>  ALUOUT <= ABUS and BBUS; -- AND
 when "0011" =>  ALUOUT <= ABUS or BBUS; -- OR
 when "0100" =>  ALUOUT <= ABUS xor BBUS; -- XOR
 when "0101" =>  ALUOUT <= not ABUS; -- NOT
 when "0110" =>  ALUOUT <= ABUS; -- MOVE
 when others => ALUOUT <= tmp_out1;
 end case;
end process;

end Behavioral;
```

Testbench VHDL code for 16-bit ALU:

```
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for 16-bit ALU
-- Testbench VHDL code for 16-bit ALU
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
use IEEE.std_logic_unsigned.all;
-- Testbench
ENTITY tb_ALU IS
END tb_ALU;

ARCHITECTURE behavior OF tb_ALU IS

    -- Component Declaration for the 16-bit ALU

    COMPONENT ALU
    PORT(
         ABUS : IN  std_logic_vector(15 downto 0);
         BBUS : IN  std_logic_vector(15 downto 0);
         ALUctrl : IN  std_logic_vector(3 downto 0);
         ALUOUT : OUT  std_logic_vector(15 downto 0)
        );
    END COMPONENT;

   --Inputs
   signal ABUS : std_logic_vector(15 downto 0) := (others => '0');
   signal BBUS : std_logic_vector(15 downto 0) := (others => '0');
   signal ALUctrl : std_logic_vector(3 downto 0) := (others => '0');

  --Outputs
   signal ALUOUT : std_logic_vector(15 downto 0);

BEGIN

 -- Instantiate the 16-bit ALU
```

```
   uut: ALU PORT MAP (
          ABUS => ABUS,
          BBUS => BBUS,
          ALUctrl => ALUctrl,
          ALUOUT => ALUOUT
        );
   stim_proc: process
   begin
      ABUS <= x"000A";
  BBUS <= x"0002";
  ALUctrl <= x"0";
  -- change ALU Control input
   for i in 0 to 15 loop
  ALUctrl <= ALUctrl + x"1";
  wait for 100 ns;
   end loop;
      ABUS <= x"00F6";
  BBUS <= x"000A";
      wait;
   end process;

END;
```

Simulation Waveform for 16-bit ALU:



The simulation waveform shows the correct operations of the 16-bit ALU. After fully verifying the VHDL 16-bit ALU, the 16-bit ALU will be used in the processing unit of the co-processor in the next post.

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